

Chapter 3

PAIRS OF RANDOM VARIABLES

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LEARNING OBJECTIVES



After careful study of this chapter you should be able to do the following:

- 1 Use joint probability mass functions and joint probability density functions to calculate probabilities.
- 2 Calculate marginal and conditional probability distributions from joint probability distributions.
- 3 Interpret and calculate covariances and correlations between random variables.
- 4 Use the multinomial distribution to determine probabilities.
- 5 Understand properties of a bivariate normal distribution and be able to draw contour plots for the probability density function.
- 6 Calculate means and variances for linear combinations of random variables and calculate probabilities for linear combinations of normally distributed random variables
- 7 Understand properties of central limit theorem and applications of the central limit theorem.

PAIRS OF RANDOM VARIABLES



- In Chapter 2 we studied probability distributions for a single random variable. However, it is often useful to have more than one random variable defined in a random experiment. For example, in the classification of transmitted and received signals, each signal can be classified as high, medium, or low quality. We might define the random variable X to be the number of high-quality signals received and the random variable Y to be the number of low-quality signals received.
- If X and Y are two random variables, the probability distribution that defines their simultaneous behavior is called a joint probability distribution. In this chapter, we investigate some important properties of these joint distributions.



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Joint Probability Mass Function

- If X and Y are discrete random variables, the joint probability distribution of X and Y is a description of the set of points (x, y) in the range of (X, Y) along with the probability of each point.
- The joint probability distribution of two random variables is sometimes referred to as the bivariate probability distribution or bivariate distribution of the random variables.
- One way to describe the joint probability distribution of two discrete random variables is through a joint probability mass function.
- Also, $P(X = x \text{ and } Y = y)$ is usually written as $P(X = x, Y = y)$.



Joint Probability Mass Function

Definition 1 (Joint Probability Mass Function)

The function $P_{X,Y}(x,y)$ is a joint probability distribution or probability mass function (PMF) of the discrete random variables X and Y if

- (a) $P_{X,Y}(x,y) \geq 0$ for all $(x,y) \in \mathbb{R}^2$.
- (b) $\sum_{x \in S_X} \sum_{y \in S_Y} P_{X,Y}(x,y) = 1$.
- (c) $P_{X,Y}(x,y) = P(X=x, Y=y)$.

✎ We use the notation $S_{X,Y}$ to denote the set of possible values of the pair (X,Y) . That is,

$$S_{X,Y} = \{(x,y) \mid P_{X,Y}(x,y) > 0\}. \quad (1)$$



Joint Probability Mass Function

Theorem 1

For all $A \subset S_{X,Y}$,

$$P[(X, Y) \in A] = \sum \sum_{(x,y) \in A} P_{X,Y}(x, y). \quad (2)$$



Joint Probability Mass Function

✍ There are various ways to represent a joint PMF. We use three of them in the following example: a list, a matrix, and a graph.

Example 1

Three ballpoint pens are selected at random from a box that contains 5 blue pens, 4 red pens, and 3 green pens. If X is the number of blue pens selected and Y is the number of red pens selected, find (a) the joint probability function $P_{X,Y}(x,y)$. (b) $P[(X,Y) \in A]$, where A is the region $\{(x,y) \mid x+y \leq 1\}$.



Joint Probability Mass Function

Solution.

- (a) The possible pairs of values (x, y) are $(0, 0)$; $(0, 1)$; $(0, 2)$; $(0, 3)$; $(1, 0)$; $(1, 1)$; $(1, 2)$; $(2, 0)$; $(2, 1)$; and $(3, 0)$. Now, for $x = 0, 1, 2, 3$; $y = 0, 1, 2, 3$, and $x + y \leq 3$,

$$P(X = x, Y = y) = \frac{(C_5^x)(C_4^y)(C_3^{3-x-y})}{C_{12}^3}.$$

A representation of $P_{X,Y}(x, y)$ is the matrix:

$P_{X,Y}(x, y)$	$y = 0$	$y = 1$	$y = 2$	$y = 3$
$x = 0$	1/220	3/55	9/110	1/55
$x = 1$	3/44	3/11	3/22	0
$x = 2$	3/22	2/11	0	0
$x = 3$	1/22	0	0	0

Table 1: Joint probability distribution in Example 1(a)

- (b) The probability that (X, Y) fall in the region A is

$$P[(X, Y) \in A] = P_{X,Y}(0, 0) + P_{X,Y}(0, 1) + P_{X,Y}(1, 0) = \frac{1}{220} + \frac{3}{55} + \frac{3}{44} = \frac{7}{55}.$$



Joint Probability Mass Function

Example 2

Test two integrated circuits one after the other. On each test, the possible outcomes are A (accept) and R (reject). Assume that all circuits are acceptable with probability 0.9 and that the outcomes of successive tests are independent. Count the number of acceptable circuits X and count the number of successful tests Y before you observe the first reject. (If both tests are successful, let $Y = 2$.) (a) Find the joint PMF of X and Y . (b) Find the probability of the event B that X , the number of acceptable circuits, equals Y , the number of tests before observing the first failure.



Joint Probability Mass Function

Solution.

(a) The sample space of the experiment is $S_{X,Y} = \{AA, AR, RA, RR\}$. We compute

$$P(AA) = 0.81, \quad P(AR) = P(RA) = 0.09, \quad P(RR) = 0.01.$$

Each outcome specifies a pair of values X and Y . Let $g(s)$ be the function that transforms each outcome s in the sample space S into the pair of random variables (X, Y) . Then

$$g(AA) = (2, 2), \quad g(AR) = (1, 1), \quad g(RA) = (1, 0), \quad g(RR) = (0, 0).$$

For each pair of values (x, y) , $P_{X,Y}(x, y)$ is the sum of the probabilities of the outcomes for which $X = x$ and $Y = y$. For example, $P_{X,Y}(1, 1) = P(AR)$. The joint PMF can be given as a set of labeled points in the (x, y) plane where each point is a possible value (probability > 0) of the pair (x, y) , or as a simple list:

$$P_{X,Y}(x, y) = \begin{cases} 0.81, & x = 2, y = 2, \\ 0.09, & x = 1, y = 1, \\ 0.09, & x = 1, y = 0, \\ 0.01, & x = 0, y = 0, \\ 0, & \text{otherwise.} \end{cases}$$



Joint Probability Mass Function

Solution.

(b) Mathematically, B is the event $(X = Y)$. The elements of B with nonzero probability are

$$B \cap S_{X,Y} = \{(0, 0), (1, 1), (2, 2)\}.$$

Therefore,

$$P(B) = P_{X,Y}(0, 0) + P_{X,Y}(1, 1) + P_{X,Y}(2, 2) = 0.01 + 0.09 + 0.81 = 0.91.$$



Joint Cumulative Distribution Function

Definition 2 (Joint Cumulative Distribution Function)

The joint cumulative distribution function (CDF) of random variables X and Y is

$$F_{X,Y}(x,y) = P(X < x, Y < y), \quad x, y \in \mathbb{R}. \quad (3)$$



Joint Cumulative Distribution Function

Theorem 2

For any pair of random variables, X, Y ,

- (a) $0 \leq F_{X,Y}(x, y) \leq 1$,
- (b) $F_{X,Y}(-\infty, y) = F_{X,Y}(x, -\infty) = 0$,
- (c) If $x < x_1$ and $y < y_1$, then $F_{X,Y}(x, y) \leq F_{X,Y}(x_1, y_1)$,
- (d) $F_{X,Y}(+\infty, +\infty) = 1$,
- (e) If $x_1 < x_2$, $y_1 < y_2$, then

$$P[x_1 \leq X < x_2, y_1 \leq Y < y_2] = F_{X,Y}(x_2, y_2) - F_{X,Y}(x_2, y_1) - F_{X,Y}(x_1, y_2) + F_{X,Y}(x_1, y_1).$$



Joint Probability Density Functionn

Definition 3 (Joint Probability Density Function)

Given the joint CDF of random variables X and Y . The joint probability density function (PDF) of the continuous random variables X and Y is a function $f_{X,Y}(x, y)$ with the property

$$F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u, v) dv du. \quad (4)$$



Joint Probability Density Function

Given $F_{X,Y}(x,y)$, Definition 3 implies that $f_{X,Y}(x,y)$ is a derivative of the CDF.

Theorem 3

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x \partial y}. \quad (5)$$



Joint Probability Density Function

✎ Properties (c) and (d) of Theorem 2 for the CDF $F_{X,Y}(x, y)$ imply corresponding properties for the PDF.

Theorem 4

A joint PDF $f_{X,Y}(x, y)$ has the following properties

(a) $f_{X,Y}(x, y) \geq 0$ for all (x, y) .

(b)
$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{X,Y}(x, y) dx dy = 1.$$

Theorem 5

The probability that the continuous random variables (X, Y) are in A is

$$P(A) = \iint_A f_{X,Y}(x, y) dx dy. \quad (6)$$



Joint Probability Density Function

Example 3

Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} k, & 0 \leq x \leq 6, 0 \leq y \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Find (a) the constant k and (b) $P(A) = P(2 \leq x < 4; 2 \leq y < 3)$.

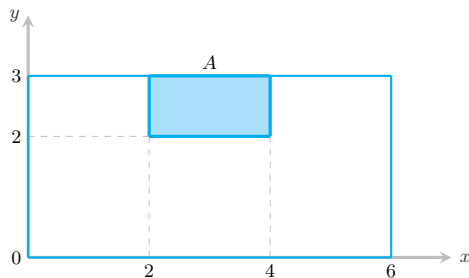
Solution. (a) The large rectangle in the diagram is the area of nonzero probability. Theorem 4 states that the integral of the joint PDF over this rectangle is 1:

$$1 = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dx dy = \int_0^6 dx \int_0^3 k dy = 18k.$$

Therefore, $k = 1/18$.



Joint Probability Density Function



(b) The small cyan rectangle in the diagram is the event $A = (2 \leq X < 4, 2 \leq Y < 3)$. $P(A)$ is the integral of the PDF over this rectangle, which is

$$P[(X, Y) \in A] = P(2 \leq X < 4; 2 \leq Y < 3)$$

$$= \int_2^4 dx \int_2^3 \frac{1}{18} dy = \frac{1}{9}.$$

Probability Distribution of a Function of Two Random Variables



We perform an experiment and observe sample values of two random variables X and Y . Based on our knowledge of the experiment, we have a probability model for X and Y embodied in the joint PMF $P_{X,Y}(x,y)$ or a joint PDF $f_{X,Y}(x,y)$. After performing the experiment, we calculate a sample value of the random variable $W = g(X,Y)$. The mathematical problem is to derive a probability model for W .

- When X and Y are discrete random variables, S_W , the range of W , is a countable set corresponding to all possible values of $g(X,Y)$. Therefore, W is a discrete random variable and has a PMF $P_W(w)$.
- When X and Y are continuous random variables and $g(x,y)$ is a continuous function, $W = g(X,Y)$ is a continuous random variable. To find the PDF, $f_W(w)$, it is usually helpful to first find the CDF $F_W(w) = P(W < w)$ and then calculate the derivative.



Probability Distribution of a Function of Two Random Variables

Example 4

In Example 3, let $W = \max\{X, Y\}$. Find the PDF $f_W(w)$ of W .

Solution. We have

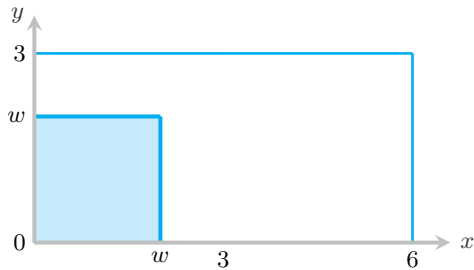
$$F_W(w) = P[\max(X, Y) < w] = P(X < w, Y < w) = \iint_A \frac{1}{18} dx dy,$$

where, $A = \{(x, y) \in \mathbb{R}^2 \mid (x < w, y < w)\}$.

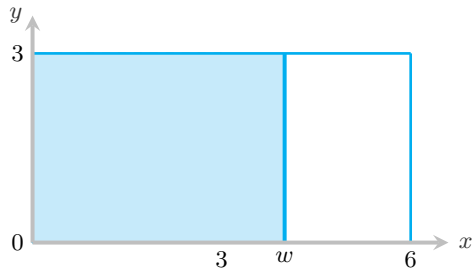
- If $w \leq 0$, $F_W(w) = 0$.
- If $0 < w \leq 3$, $F_W(w) = \int_0^w dx \int_0^w \frac{1}{18} dy = \frac{w^2}{18}$.
- If $3 < w \leq 6$, $F_W(w) = \int_0^w dx \int_0^3 \frac{1}{18} dy = \frac{w}{6}$.



Probability Distribution of a Function of Two Random Variables



(a)



(b)

Figure 1: (a) Domain $(0 < w \leq 3)$; (b) Domain $(3 < w \leq 6)$



Joint Cumulative Distribution Function

Solution. If $w > 6$, $F_W(w) = 1$.

Hence,

$$F_W(w) = \begin{cases} 0 & \text{if } w \leq 0, \\ \frac{w^2}{18} & \text{if } 0 < w \leq 3, \\ \frac{w}{6} & \text{if } 3 < w \leq 6, \\ 1 & \text{if } w > 6. \end{cases}$$

Therefore,

$$f_W(w) = \begin{cases} \frac{w}{9} & \text{if } 0 < w \leq 3, \\ \frac{1}{6} & \text{if } 3 < w \leq 6, \\ 0 & \text{if } w \leq 0, w > 6. \end{cases}$$



Expected Values of a Function of Two Random Variables

✎ There are many situations in which we are interested only in the expected value of a derived random variable $W = g(X, Y)$, not the entire probability model. In these situations, we can obtain the expected value directly from $P_{X,Y}(x, y)$ or $f_{X,Y}(x, y)$ without taking the trouble to compute $P_W(w)$ or $f_W(w)$.

Theorem 6

For random variables X and Y , the expected value of $W = g(X, Y)$ is

$$E(W) = \sum_{x \in S_X} \sum_{y \in S_Y} g(x, y) P_{X,Y}(x, y) \quad (7)$$

if W is a discrete random variable and

$$E(W) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) f_{X,Y}(x, y) dx dy \quad (8)$$

if W is a continuous random variable.



Expected Values of a Function of Two Random Variables

Example 5

For X and Y in Example 1, using (7),

$$E(XY) = (1)(1)(3/11) + (1)(2)(3/2) + (2)(1)(2/11) = 40/11.$$

Example 6

For X and Y in Example 3, using (8),

$$E(XY) = \int_0^6 \left(\int_0^3 xy \frac{1}{18} dy \right) dx = \int_0^6 \frac{1}{4} x dx = \frac{36}{8} = 4.5.$$

Expected Values of a Function of Two Random Variables



Theorem 7

$$E[g_1(X, Y) + \dots + g_n(X, Y)] = E[g_1(X, Y)] + \dots + E[g_n(X, Y)]. \quad (9)$$

Theorem 8

For any two random variables X and Y ,

$$E(X + Y) = E(X) + E(Y). \quad (10)$$



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Marginal Probability Mass Function

Theorem 9

For discrete random variables X and Y with joint PMF $P_{X,Y}(x,y)$, the marginal PMF of X and Y are

$$P_X(x) = \sum_{y \in S_Y} P_{X,Y}(x,y) \quad \text{and} \quad P_Y(y) = \sum_{x \in S_X} P_{X,Y}(x,y). \quad (11)$$



Marginal Probability Mass Function

Example 7

In Example 2, we found the joint PMF of X and Y to be

$P_{X,Y}(x,y)$	$y = 0$	$y = 1$	$y = 2$
$x = 0$	0.01	0	0
$x = 1$	0.09	0.09	0
$x = 2$	0	0	0.81

Find the marginal PMFs for the random variables X and Y .



Marginal Probability Mass Function

Solution. To find $P_X(x)$, we note that both X and Y have range $\{0, 1, 2\}$. Theorem 9 gives

$$\begin{aligned}P_X(0) &= \sum_{y=0}^2 P_{X,Y}(0, y) = 0.01, & P_X(1) &= \sum_{y=0}^2 P_{X,Y}(1, y) = 0.18 \\P_X(2) &= \sum_{y=0}^2 P_{X,Y}(2, y) = 0.81, & P_X(x) &= 0, \quad x \neq 0, 1, 2.\end{aligned}$$

For the PMF of Y , we obtain

$$\begin{aligned}P_Y(0) &= \sum_{x=0}^2 P_{X,Y}(x, 0) = 0.10, & P_Y(1) &= \sum_{x=0}^2 P_{X,Y}(x, 1) = 0.09 \\P_Y(2) &= \sum_{x=0}^2 P_{X,Y}(x, 2) = 0.81, & P_Y(y) &= 0, \quad y \neq 0, 1, 2.\end{aligned}$$



Marginal Probability Mass Function

Solution. Hence,

$$P_X(x) = \begin{cases} 0.01, & x = 0, \\ 0.18, & x = 1, \\ 0.81, & x = 2, \\ 0, & \text{otherwise.} \end{cases} \quad P_Y(y) = \begin{cases} 0.1, & y = 0, \\ 0.09, & y = 1, \\ 0.81, & y = 2, \\ 0, & \text{otherwise.} \end{cases}$$



Marginal Probability Mass Function

Example 8

Referring to the matrix representation of $P_{X,Y}(x,y)$ in Example 2, we observe that each value of $P_X(x)$ is the result of adding all the entries in one row of the matrix. Each value of $P_Y(y)$ is a column sum. We display $P_X(x)$ and $P_Y(y)$ by rewriting the matrix in Example 2 and placing the row sums and column sums in the margins.

$P_{X,Y}(x,y)$	$y = 0$	$y = 1$	$y = 2$	$y = 3$	$P_X(x)$
$x = 0$	1/220	3/55	9/110	1/55	35/220
$x = 1$	3/44	3/11	3/22	0	105/220
$x = 2$	3/22	2/11	0	0	70/220
$x = 3$	1/22	0	0	0	10/220
$P_Y(y)$	56/220	112/220	48/220	4/220	1



Marginal Probability Mass Function

Note that the sum of all the entries in the bottom margin is 1 and so is the sum of all the entries in the right margin. The complete marginal PMF, $P_Y(y)$, appears in the bottom row of the table, and $P_X(x)$ appears in the last column of the table. From the matrix of $P_{X,Y}(x, y)$, we have marginal probability distribution of X is

X	0	1	2	3
$P(X = x_i)$	7/44	21/44	7/22	1/22

and marginal probability distribution of Y is

Y	0	1	2	3
$P(Y = y_j)$	14/55	28/55	12/55	1/55



Marginal Probability Mass Function

Remark 1

The expected value $E(X)$ and variance $\text{Var}(X)$ can be obtained by first calculating the marginal probability distribution of X and then determining $E(X)$ and $\text{Var}(X)$ by the usual method.

Example 9

Find the expected value $E(X)$ and variance $\text{Var}(X)$ of random variable X in Example 2.

Solution. Using the result of Example 7,

$$E(X) = (0)(0.01) + (1)(0.18) + (2)(0.81) = 1.8,$$

$$\text{Var}(X) = (0^2)(0.01) + (1^2)(0.18) + (2^2)(0.81) - (1.8)^2 = 0.18.$$



Marginal Cumulative Distribution Functions

The marginal CDF has properties that are direct consequences of the definition. For example, we note that the event $(X < x)$ suggests that Y can have any value so long as the condition on X is met. This corresponds to the joint event $(X < x, Y < \infty)$. Therefore,

$$F_X(x) = P(X < x) = P(X < x, Y < \infty) = \lim_{y \rightarrow \infty} F_{X,Y}(x, y) = F_{X,Y}(x, \infty). \quad (12)$$

Hence, the marginal cumulative distribution functions are

$$F_X(x) = F_{X,Y}(x, \infty) \quad \text{and} \quad F_Y(y) = F_{X,Y}(\infty, y). \quad (13)$$



Marginal Probability Density Function

Theorem 10

If the joint probability density function of random variables X and Y is $f_{X,Y}(x,y)$, the marginal probability density functions of X and Y are

$$f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y)dy \quad \text{and} \quad f_Y(y) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y)dx. \quad (14)$$



Marginal Probability Density Function

Example 10

The joint PDF of X and Y is

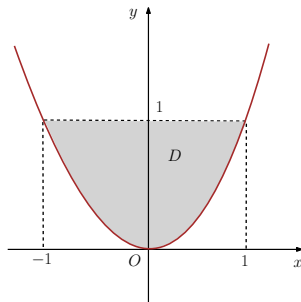
$$f_{X,Y}(x,y) = \begin{cases} \frac{5y}{4}, & -1 \leq x \leq 1, x^2 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the marginal PDFs $f_X(x)$ and $f_Y(y)$.



Marginal Probability Density Function

Solution. Set $D = \{(x, y) \in \mathbb{R}^2 \mid -1 \leq x \leq 1, x^2 \leq y \leq 1\}$.



We use Theorem 10 to

find the marginal PDF $f_X(x)$. When $x < -1$ or when $x > 1$, $f_{X,Y}(x, y) = 0$, and therefore $f_X(x) = 0$. For $-1 \leq x \leq 1$,

$$f_X(x) = \int_{x^2}^1 \frac{5y}{4} dy = \frac{5(1 - x^4)}{8}.$$

The complete expression for the marginal PDF of X is

$$f_X(x) = \begin{cases} \frac{5(1 - x^4)}{8}, & -1 \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$



Marginal Probability Density Function

Solution. For the marginal PDF of Y , we note that for $y < 0$ or $y > 1$, $f_Y(y) = 0$. For $0 \leq y \leq 1$, we integrate over the horizontal bar marked $Y = y$. The boundaries of the bar are $x = -\sqrt{y}$ and $x = \sqrt{y}$. Therefore, for $0 \leq y \leq 1$,

$$f_Y(y) = \int_{-\sqrt{y}}^{\sqrt{y}} \frac{5y}{4} dx = \frac{5y}{4} x \Big|_{x=-\sqrt{y}}^{x=\sqrt{y}} = \frac{5y^{3/2}}{2}.$$

The complete marginal PDF of Y is $f_Y(y) = \begin{cases} \frac{5y^{3/2}}{2}, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$



Marginal Probability Density Function

Example 11

The joint PDF of X and Y is

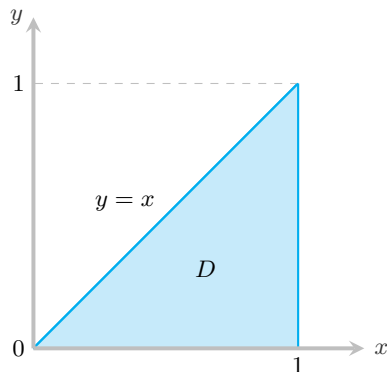
$$f(x, y) = \begin{cases} kx, & \text{if } 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find constant k . (b) Find the marginal PDFs $f_X(x)$ and $f_Y(y)$. (c) Find the expected value and variance of X .



Marginal Probability Density Function

Solution. Domain $D = \{(x, y) \mid 0 < y < x < 1\}$.



(a) From Theorem 4, $k \geq 0$ and

$$\begin{aligned} 1 &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{X,Y}(x, y) dx dy = \iint_D kx dx dy = \int_0^1 dx \int_0^x kx dy \\ &= k \int_0^1 x^2 dx = \frac{k}{3}. \end{aligned}$$

Hence $k = 3$ and

$$f(x, y) = \begin{cases} 3x, & \text{if } (x, y) \in D, \\ 0, & \text{otherwise.} \end{cases}$$



Marginal Probability Density Function

Solution. (b) From Theorem 10,

$$f_X(x) = \int_{-\infty}^{+\infty} f(x, y) dy = \begin{cases} \int_0^x 3x dy, & 0 < x < 1, \\ 0, & \text{otherwise,} \end{cases} = \begin{cases} 3x^2, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_y^1 3x dx, & 0 < y < 1, \\ 0, & \text{otherwise,} \end{cases} = \begin{cases} \frac{3}{2} - \frac{3}{2}y^2, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(c) We have

$$E(X) = \int_0^1 x \times 3x^2 dx = \frac{3}{4} \quad \text{and} \quad \text{Var}(X) = E(X^2) - [E(X)]^2 = \int_0^1 x^2 \times 3x^2 dx - \left(\frac{3}{4}\right)^2 = \frac{3}{80}.$$



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Conditional Joint PMF

Definition 4 (Conditional Joint PMF)

For discrete random variables X and Y and an event, B with $P(B) > 0$, the conditional joint PMF of X and Y given B is

$$P_{X,Y|B}(x,y) = P(X = x, Y = y|B). \quad (15)$$

Theorem 11

For any event B , a region of the X, Y plane with $P(B) > 0$,

$$P_{X,Y|B}(x,y) = \begin{cases} \frac{P_{X,Y}(x,y)}{P(B)}, & (x,y) \in B, \\ 0, & \text{otherwise.} \end{cases} \quad (16)$$



Conditional Joint PDF

Definition 5 (Conditional Joint PDF)

Given an event B with $P(B) > 0$, the conditional joint probability density function of X and Y is

$$f_{X,Y|B}(x,y) = \begin{cases} \frac{f_{X,Y}(x,y)}{P(B)}, & (x,y) \in B, \\ 0, & \text{otherwise.} \end{cases} \quad (17)$$

Example 12

X and Y are random variables with joint PDF

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{15}, & 0 \leq x \leq 5, 0 \leq y \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Find the conditional PDF of X and Y given the event $B = \{X + Y \geq 4\}$.



Conditional Joint PDF

Solution. We calculate $P(B)$ by integrating $f_{X,Y}(x,y)$ over the region B .

$$P(B) = \int_0^3 \int_{4-y}^5 \frac{1}{15} dx dy = \frac{1}{15} \int_0^3 (1+y) dy = \frac{1}{2}.$$

Definition 5 leads to the conditional joint PDF

$$f_{X,Y|B}(x,y) = \begin{cases} \frac{2}{15}, & 0 \leq x \leq 5, 0 \leq y \leq 3, x+y \geq 4 \\ 0, & \text{otherwise.} \end{cases}$$



Conditional Expected Value and Variance

Theorem 12 (Conditional Expected Value)

For random variables X and Y and an event B of nonzero probability, the conditional expected value of $W = g(X, Y)$ given B is

$$E(W|B) = \sum_{x \in S_X} \sum_{y \in S_Y} g(x, y) P_{X,Y|B}(x, y) \quad (18)$$

if W is a discrete random variable and

$$E(W|B) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(x, y) f_{X,Y|B}(x, y) dx dy \quad (19)$$

if W is a continuous random variable.



Conditional Expected Value and Variance

Definition 6 (Conditional Variance)

The conditional variance of the random variable $W = g(X, Y)$ is

$$\text{Var}(W|B) = E[(W - E(W|B))^2|B]. \quad (20)$$

Theorem 13

$$\text{Var}(W|B) = E(W^2|B) - [E(W|B)]^2. \quad (21)$$



Conditional Expected Value and Variance

Example 13

Continuing Example 12, find the conditional expected value of $W = XY$ given the event $B = \{X + Y \geq 4\}$.

Solution. From Theorem 12,

$$E(XY|B) = \int_0^3 \int_{4-y}^5 \frac{2}{15} xy dx dy = \frac{1}{15} \int_0^3 x^2 \Big|_{4-y}^5 y dy = \frac{1}{15} \int_0^3 (9y + 8y^2 - y^3) dy = \frac{123}{20}.$$



Conditional PMF

Definition 7 (Conditional PMF)

For any event $Y = y$ such that $P_Y(y) > 0$, the conditional PMF of X given $Y = y$ is

$$P_{X|y} := P_{X|(Y=y)}(x) = P[(X = x)|(Y = y)], \quad (22)$$

where

$$P[(X = x)|(Y = y)] = \frac{P(X = x, Y = y)}{P(Y = y)}.$$



Conditional PMF

✎ The following theorem contains the relationship between the joint PMF of X and Y and the two conditional PMFs, $P_{X|Y}(x)$ and $P_{Y|X}(y)$.

Theorem 14

For random variables X and Y with joint PMF $P_{X,Y}(x,y)$, and x and y such that $P_X(x) > 0$ and $P_Y(y) > 0$,

$$P_{X,Y}(x,y) = P_{X|(Y=y)}(x)P_Y(y) = P_{Y|(X=x)}(y)P_X(x). \quad (23)$$



Conditional PMF

Example 14

In Example 1,

$$P(X = 0|Y = 1) = \frac{P(X = 0, Y = 1)}{P(Y = 1)} = \frac{3}{28}, \quad P(X = 1|Y = 1) = \frac{P(X = 1, Y = 1)}{P(Y = 1)} = \frac{15}{28},$$

$$P(X = 2|Y = 1) = \frac{P(X = 2, Y = 1)}{P(Y = 1)} = \frac{10}{28}, \quad P(X = 3|Y = 1) = \frac{P(X = 3, Y = 1)}{P(Y = 1)} = 0,$$

therefore, the conditional probability distribution of X given $Y = 1$ is

$X (Y = 1)$	0	1	2	3
P	3/28	15/28	5/14	0



Conditional PMF

Theorem 15 (Conditional Expected Value of a Function)

Let X and Y are discrete random variables. For any $y \in S_Y$, the conditional expected value of $g(X, Y)$ given $(Y = y)$ is

$$E[g(X, Y)|Y = y] = \sum_{x \in S_X} g(x, y)P_{X|y}(x). \quad (24)$$

The conditional expected value of X given $Y = y$ is a special case of Theorem 15:

$$E([X|Y = y]) = \sum_{x \in S_X} xP_{X|y}(x). \quad (25)$$



Conditional PMF

Example 15

Returning to Example 14,

$$E(X|Y = 1) = 0 \times \frac{3}{28} + 1 \times \frac{15}{28} + 2 \times \frac{5}{14} + 3 \times 0 = 1.25.$$



Conditional PDF

✎ Now we consider the case in which X and Y are continuous random variables. We observe $(Y = y)$ and define the PDF of X given $(Y = y)$. We cannot use $B = (Y = y)$ in Definition 5 because $P(Y = y) = 0$. Instead, we define a conditional probability density function, denoted as $f_{X|y}(x) : f_{X|(Y=y)}(x)$.

Definition 8 (Conditional PDF)

For y such that $f_Y(y) > 0$, the conditional PDF of X given $(Y = y)$ is

$$f_{X|y}(x) := f_{X|(Y=y)}(x) = \frac{f_{X,Y}(x, y)}{f_Y(y)}. \quad (26)$$



Conditional PDF

✎ Definition 8 implies

$$f_{Y|x}(y) := f_{Y|(X=x)}(y) = \frac{f_{X,Y}(x,y)}{f_X(x)}, \quad f_X(x) > 0. \quad (27)$$

Remark 2

Because the conditional probability density function $f_{Y|(X=x)}(y)$ is a probability density function for all y , the following properties are satisfied:

- (a) $f_{Y|x}(y) \geq 0$.
- (b) $\int_{-\infty}^{+\infty} f_{Y|x}(y) dy = 1$.
- (c) $P(Y \in B | (X = x)) = \int_B f_{Y|x}(y) dy$ for any set B in the range of Y .



Conditional PDF

Example 16

The joint probability density function of random variables X and Y is

$$f_{X,Y}(x,y) = \begin{cases} 2, & 0 < y \leq x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) For $0 < x < 1$, find the conditional PDF $f_{Y|x}(y)$. (b) For $0 < y < 1$, find the conditional PDF $f_{X|y}(x)$.

Solution. (a) For $0 < x < 1$, Theorem 10 implies $f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y)dy = \int_0^x 2dy = 2x$. The conditional PDF of Y given X is

$$f_{Y|x}(y) = \frac{f_{X,Y}(x,y)}{f_X(x)} = \begin{cases} 1/x, & 0 < y < x, \\ 0, & \text{otherwise.} \end{cases}$$

Given $X = x$, we see that Y is the uniform $(0, x)$ random variable.



Conditional PDF

Solution. (b) For $0 < y < 1$, Theorem 10 implies

$$f_Y(y) = \int_{-\infty}^{+\infty} f_{X,Y}(x,y)dx = \int_y^1 2dx = 2(1-y).$$

Furthermore, Equation (27) implies

$$f_{X|Y}(x) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \begin{cases} 1/(1-y), & y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Conditioned on $Y = y$, we see that X is the uniform $(y, 1)$ random variable.



Conditional PDF

Example 17

The joint probability density function of random variables X and Y is

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{x}, & \text{if } 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Find the PDFs $f_X(x)$, $f_Y(y)$. (b) Find the conditional PDFs $f_{X|y}(x)$, $f_{Y|x}(y)$.

Solution. (a) The PDFs of X and Y are:

$$f_X(x) = \int_{-\infty}^{+\infty} f(x,y)dy = \begin{cases} \int_0^x \frac{1}{x} dy, & 0 < x < 1, \\ 0, & \text{otherwise,} \end{cases} = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$



Conditional PDF

Solution.

$$f_Y(y) = \int_{-\infty}^{+\infty} f(x, y) dx = \begin{cases} \int_y^1 \frac{1}{x} dx, & 0 < y < 1, \\ 0, & \text{otherwise,} \end{cases} = \begin{cases} -\ln y, & 0 < y < 1, \\ 0, & \text{otherwise.} \end{cases}$$

(b) The conditional PDFs are

$$f_{X|Y}(x) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \begin{cases} -\frac{1}{x \ln y}, & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

$$f_{Y|X}(y) = \frac{f_{X,Y}(x, y)}{f_X(x)} = \begin{cases} \frac{1}{x}, & 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Conditional PDF



Theorem 16

$$f_{X,Y}(x,y) = f_{Y|x}(y)f_X(x) = f_{X|y}(x)f_Y(y). \quad (28)$$



Conditional PDF

Definition 9 (Conditional Expected Value of a Function)

For continuous random variables X and Y and any y such that $f_Y(y) > 0$, the conditional expected value of $g(X, Y)$ given $Y = y$ is

$$E[g(X, Y)|Y = y] = \int_{-\infty}^{+\infty} g(x, y) f_{X|y}(x) dx. \quad (29)$$

 The conditional expected value of X given $(Y = y)$ is a special case of Definition 9:

$$E(X|Y = y) = \int_{-\infty}^{+\infty} x f_{X|y}(x) dx. \quad (30)$$



Conditional Joint PMF

Definition 10 (Conditional Expected Value)

The conditional expected value $E(X|Y)$ is a function of random variable Y such that if $(Y = y)$ then $E(X|Y) = E(X|Y = y)$.

Example 18

For random variables X and Y in Example 16 we found that the conditional PDF of X given $(Y = y)$ is

$$f_{X|Y}(x) = \frac{f_{X,Y}(x, y)}{f_Y(y)} = \begin{cases} 1/(1 - y), & y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the conditional expected values $E(X|Y = y)$ and $E(X|Y)$.



Conditional Joint PMF

Solution. Given the conditional PDF $f_{X|Y}(x)$, we perform the integration

$$E(X|Y = y) = \int_{-\infty}^{+\infty} x f_{X|Y}(x) dx = \int_y^1 \frac{1}{1-y} x dx = \frac{x^2}{2(1-y)} \Big|_{x=y}^{x=1} = \frac{1+y}{2}.$$

Since $E(X|Y = y) = (1+y)/2$, $E(X|Y) = (1+Y)/2$.



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Independence

Definition 11 (Independent Random Variables)

Discrete random variables X and Y are independent if and only if

$$P_{X,Y}(x,y) = P_X(x)P_Y(y). \quad (31)$$

Continuous random variables X and Y are independent if and only if

$$f_{X,Y}(x,y) = f_X(x)f_Y(y). \quad (32)$$



Independence

✎ Theorem 14 implies that if X and Y are independent discrete random variables, then

$$P_{X|y}(x) = P_X(x), \quad P_{Y|x}(y) = P_Y(y). \quad (33)$$

Theorem 16 implies that if X and Y are independent continuous random variables, then

$$f_{X|y}(x) = f_X(x), \quad f_{Y|x}(y) = f_Y(y). \quad (34)$$



Independence

Example 19

$$f_{X,Y}(x,y) = \begin{cases} 4xy, & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Are X and Y independent?

Solution. The marginal PDFs of X and Y are

$$f_X(x) = \begin{cases} 2x, & 0 \leq x \leq 1, \\ 0, & \text{otherwise,} \end{cases} \quad f_Y(y) = \begin{cases} 2y, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

It is easily verified that $f_{X,Y}(x,y) = f_X(x)f_Y(y)$ for all pairs (x,y) and so we conclude that X and Y are independent.



Independence

Theorem 17

For independent random variables X and Y ,

- (a) $E[g(X)h(Y)] = E[g(X)]E[h(Y)],$
- (b) $E(XY) = E(X)E(Y),$
- (c) $E[X|(Y = y)] = E(X)$ for all $y \in S_Y,$
- (d) $E[Y|(X = x)] = E(Y)$ for all $x \in S_X.$
- (e) $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$ for any sets A and B in the range of X and Y , respectively.



Independence

Example 20

Let the random variables X and Y denote the lengths of two dimensions of a machined part, respectively. Assume that X and Y are independent random variables, and further assume that the distribution of X is normal with mean 10.5 millimeters and variance $0.0025 \text{ (millimeter)}^2$ and that the distribution of Y is normal with mean 3.2 millimeters and variance $0.0036 \text{ (millimeter)}^2$. Determine the probability that $10.4 < X < 10.6$ and $3.15 < Y < 3.25$.



Independence

Solution. Because X and Y are independent,

$$\begin{aligned}
 P(10.4 < X < 10.6, 3.15 < Y < 3.25) &= P(10.4 < X < 10.6)P(3.15 < Y < 3.25) \\
 &= \left[\Phi\left(\frac{10.6 - 10.5}{0.05}\right) - \Phi\left(\frac{10.4 - 10.5}{0.05}\right) \right] \times \left[\Phi\left(\frac{3.25 - 3.2}{0.06}\right) - \Phi\left(\frac{3.15 - 3.2}{0.06}\right) \right] \\
 &= \left(\Phi(2) - \Phi(-2) \right) \left(\Phi(0.83) - \Phi(-0.83) \right) \\
 &= (2 \times 0.97725 - 1)(2 \times 0.79673 - 1) = 0.5665.
 \end{aligned}$$



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PROBABILITY MODELS OF A n RANDOM VARIABLES



Read-only



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Covariance



Definition 12 (Covariance)

The covariance between the random variables X and Y , denoted as $\text{cov}(X, Y)$ or σ_{XY} is

$$\text{cov}(X, Y) = E[(X - E(X))(Y - E(Y))]. \quad (35)$$



Covariance

✎ If the points in the joint probability distribution of X and Y that receive positive probability tend to fall along a line of positive (or negative) slope, $\text{cov}(X, Y)$ is positive (or negative).

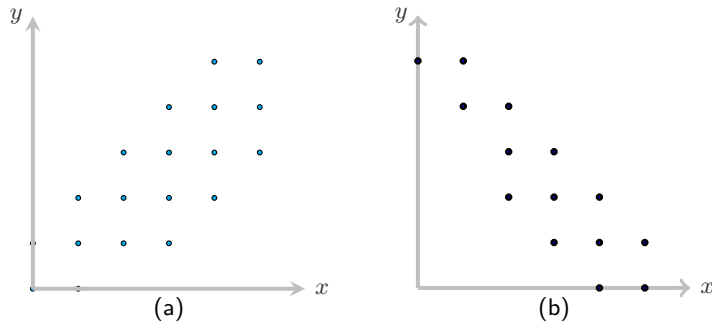


Figure 2: Joint probability distributions and the sign of covariance between X and Y (a) Positive covariance; (b) Negative covariance

Covariance

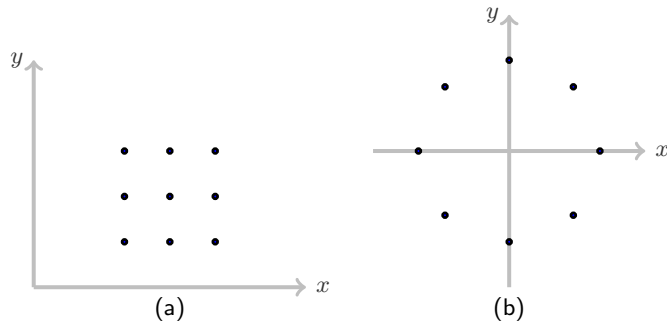


Figure 3: Joint probability distributions and the sign of covariance between X and Y : Zero covariance

Covariance



Example 21

In Example 1, the random variables X and Y are the number of blue pens and red pens selected, respectively. Is the covariance between X and Y positive or negative?

Solution. As the number of blue pens increase, the number of red pens decreases. Therefore, X and Y have a negative covariance. This can be verified from the joint probability distribution in Table 1.

Covariance



✍ The equality of the two expressions for covariance in Equation (35) is shown.

Theorem 18

$$\text{cov}(X, Y) = E(XY) - E(X)E(Y).$$

(36)



Covariance

Example 22

For the integrated circuits tests in Example 2, we found in Example 7 that the probability model for X and Y is given by the following matrix.

$P_{X,Y}(x, y)$	$y = 0$	$y = 1$	$y = 2$	$P_X(x)$
$x = 0$	0.01	0	0	0.01
$x = 1$	0.09	0.09	0	0.18
$x = 2$	0	0	0.81	0.81
$P_Y(y)$	0.10	0.09	0.81	

Find $\text{cov}(X, Y)$.

Covariance



Solution. By (7),

$$E(XY) = \sum_{x=0}^2 \sum_{y=0}^2 xy P_{X,Y}(x, y) = 1 \times 1 \times 0.09 + 2 \times 2 \times 0.81 = 3.33.$$

To use Theorem 18 to find the covariance, we find

$$E(X) = 1 \times 0.18 + 2 \times 0.81 = 1.80, \quad E(Y) = 1 \times 0.09 + 2 \times 0.81 = 1.71.$$

Therefore, by Theorem 18,

$$\text{cov}(X, Y) = 3.33 - 1.80 \times 1.71 = 0.252.$$

Covariance



Theorem 19

- (a) $\text{cov}(X, Y) = \text{cov}(Y, X)$.
- (b) $\text{Var}(X) = \text{cov}(X, X)$ and $\text{Var}(Y) = \text{cov}(Y, Y)$.
- (c) $\text{cov}(aX, Y) = a\text{cov}(X, Y)$.
- (d) $\text{cov}(X + Z, Y) = \text{cov}(X, Y) + \text{cov}(Z, Y)$.

Correlation



✎ There is another measure of the relationship between two random variables that is often easier to interpret than the covariance.

Definition 13 (Correlation)

The correlation between random variables X and Y , denoted as ρ_{XY} , is

$$\rho_{XY} = \frac{\text{cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}. \quad (37)$$



Correlation

Example 23

The joint PDF of two continuous random variables X and Y is

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{16}xy, & 0 \leq x \leq 2, 0 \leq y \leq 4, \\ 0, & \text{otherwise.} \end{cases}$$

Find ρ_{XY} .

Solution. We have

$$E(X) = \int_0^2 \left(\frac{1}{16} \int_0^4 x^2 y dy \right) dx = \frac{4}{3}, \quad E(Y) = \int_0^2 \left(\int_0^4 xy^2 dy \right) dx = \frac{8}{3}, \quad E(XY) = \int_0^2 \left(\int_0^4 x^2 y^2 dy \right) dx = \frac{32}{9}$$

and $\text{cov}(X, Y) = \frac{32}{9} - \frac{4}{3} \times \frac{8}{3} = 0$. Therefore, $\rho_{XY} = 0$.

Correlation



✎ Because $\sigma_X > 0$ and $\sigma_Y > 0$, if the covariance between X and Y is positive, negative, or zero, the correlation between X and Y is positive, negative, or zero, respectively. The following result can be shown.

Theorem 20

For any two random variables X and Y ,

$$-1 \leq \rho_{XY} \leq 1.$$

(38)

If ρ_{XY} equals $+1$ or -1 , it can be shown that the points in the joint probability distribution that receive positive probability fall exactly along a straight line. Two random variables with nonzero correlation are said to be correlated. Similar to covariance, the correlation is a measure of the linear relationship between random variables.

Correlation



Example 24

Suppose that the random variable X has the following distribution:

$$P_X(x) = \begin{cases} 0,2, & \text{if } x = 1, \\ 0,6, & \text{if } x = 2, \\ 0,2, & \text{if } x = 3, \\ 0, & \text{otherwise.} \end{cases}$$

Let $Y = 2X + 5$. That is,

$$P_Y(y) = \begin{cases} 0,2, & \text{if } y = 7, \\ 0,6, & \text{if } y = 9, \\ 0,2, & \text{if } y = 11, \\ 0, & \text{otherwise.} \end{cases}$$

Correlation

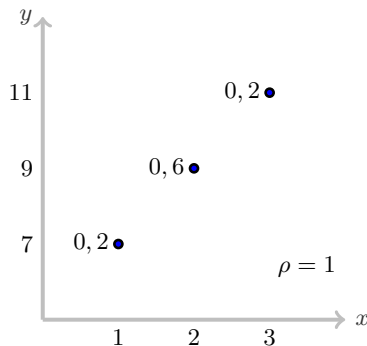


Figure 4: Joint distribution for Example 24

Refer to Fig. 4. Because X and Y are linearly related, $\rho_{XY} = 1$. This can be verified by direct calculations (37)!

Correlation



Theorem 21

If X and Y are independent random variables,

$$\text{cov}(X, Y) = \rho_{XY} = 0. \quad (39)$$

However, if the correlation between two random variables is zero, we cannot immediately conclude that the random variables are independent.

Correlation



Example 25

Random variables X and Y have a joint PMF given by the following matrix

$P_{X,Y}(x,y)$	$y = -1$	$y = 0$	$y = 1$
$x = -1$	0	0.25	0
$x = 1$	0.25	0.25	0.25

(40)

(a) Are X and Y independent? (b) Are X and Y uncorrelated?



Correlation

Solution.

(a) For the marginal PMFs, we have $P_X(-1) = 0.25$ and $P_Y(-1) = 0.25$. Thus

$$P_X(-1)P_Y(-1) = 0.0625 \neq P_{X,Y}(-1, -1) = 0,$$

and we conclude that X and Y are not independent.

(b) To find $\text{cov}(X, Y)$, we calculate

$$E(X) = 0.5, \quad E(Y) = 0, \quad E(XY) = 0.$$

Therefore,

$$\text{cov}(X, Y) = E(XY) - E(X)E(Y) = \rho_{XY} = 0,$$

and by definition X and Y are uncorrelated.

Correlation



✎ The following properties provide additional tools for calculating variance.

Theorem 22

$$\text{Var}(aX + bY + c) = a^2\text{Var}(X) + b^2\text{Var}(Y) + 2abcov(X, Y). \quad (41)$$

Corollary 1

$$\text{Var}(X \pm Y) = \text{Var}(X) + \text{Var}(Y) \pm 2cov(X, Y). \quad (42)$$



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COMMON JOINT DISTRIBUTIONS



Read-only



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LINEAR FUNCTIONS OF RANDOM VARIABLES



Read-only



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Central Limit Theorem

Theorem 23 (Central Limit Theorem)

Given $\{X_n\}_{n=1}^{\infty}$ a sequence of independent and identically distributed (iid) random variables with expected value μ and finite variance σ^2 , and if $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$, the limiting form of the distribution of

$$Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \quad (43)$$

as $n \rightarrow \infty$, is the standard normal distribution $\mathcal{N}(0, 1)$, that is

$$\lim_{n \rightarrow \infty} F_{Z_n}(z) = \Phi(z). \quad (44)$$



Central Limit Theorem

The normal approximation for depends on the sample size n . The following figures show the distributions of average scores obtained when tossing one, two, three, and five dice, respectively.



Figure 5: Distributions of average scores from throwing dice



Central Limit Theorem

To use the central limit theorem, we observe that we can express the iid sum $W_n = X_1 + X_2 + \cdots + X_n$ as

$$W_n = \sigma\sqrt{n}Z_n + n\mu. \quad (45)$$

The CDF of W_n can be expressed in terms of the CDF of Z_n as

$$F_{W_n}(w) = P(\sigma\sqrt{n}Z_n + n\mu < w) = F_{Z_n}\left(\frac{w - n\mu}{\sigma\sqrt{n}}\right). \quad (46)$$

For large n , the central limit theorem says that $F_{Z_n}(z) \simeq \Phi(z)$. This approximation is the basis for practical applications of the central limit theorem.



Central Limit Theorem

The PDF of W_n , the sum of n uniform $\mathcal{U}[0, 1]$ random variables, and the corresponding central limit theorem approximation for $n = 1, 2, 3, 4$.

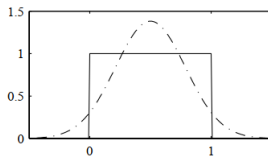


Figure 6: $n = 1$

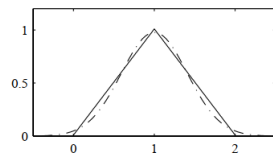


Figure 7: $n = 2$

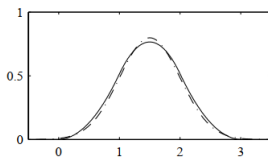


Figure 8: $n = 3$

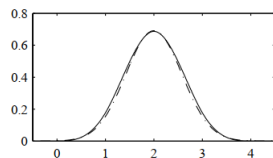


Figure 9: $n = 4$



Central Limit Theorem

Definition 14 (Central Limit Theorem Approximation)

Let $W_n = X_1 + \cdots + X_n$ be the sum of n iid random variables, each with $E(X_n) = \mu$ and $V(X_n) = \sigma^2$. The central limit theorem approximation to the CDF of W_n is

$$F_{W_n}(w) \simeq \Phi\left(\frac{w - n\mu}{\sigma\sqrt{n}}\right). \quad (47)$$



Applications of the Central Limit Theorem

Example 26

A modem transmits one million bits. Each bit is 0 or 1 independently with equal probability. (a) Estimate the probability of at least 502,000 ones. (b) Transmit one million bits. Let A denote the event that there are at least 499,000 ones but no more than 501,000 ones. What is $P(A)$?



Applications of the Central Limit Theorem

Solution. Let X_i be the value of bit i (0 or 1). Because X_i is a Bernoulli random variable with $E(X_i) = 0.5$ and $V(X_i) = 0.25$ for all i . The number of ones in one million bits is $W = \sum_{i=1}^{10^6} X_i$. W is $\mathcal{B}(n, p)$ with $n = 10^6$ and $p = 0.5$.

(a) By the central limit theorem approximation,

$$\begin{aligned} P(W \geq 502000) &= 1 - P(W < 502000) \simeq 1 - \Phi\left(\frac{502000 - 10^6 \times 0.5}{0.5\sqrt{10^6}}\right) \\ &= 1 - \Phi(4) = 1 - 0.999968 = 3.2 \times 10^{-5}. \end{aligned}$$

(b) By the central limit theorem approximation,

$$\begin{aligned} P(A) &= P(499000 < W \leq 501000) = P(W \leq 501000) - P(W < 499000) \\ &\simeq \Phi\left(\frac{501000 - 10^6 \times 0.5}{0.5\sqrt{10^6}}\right) - \Phi\left(\frac{499000 - 10^6 \times 0.5}{0.5\sqrt{10^6}}\right) \\ &= \Phi(2) - \Phi(-2) = 2 \times 0.97725 - 1 = 0.9545. \end{aligned}$$



Applications of the Central Limit Theorem

Definition 15 (De Moivre–Laplace Formula)

For a binomial $\mathcal{B}(n, p)$ random variable K ,

$$P(k_1 \leq K \leq k_2) \simeq \Phi\left(\frac{k_2 + 0.5 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{k_1 - 0.5 - np}{\sqrt{np(1-p)}}\right) \quad (48)$$



Applications of the Central Limit Theorem

To appreciate why the ± 0.5 terms increase the accuracy of approximation, consider the following simple but dramatic example in which $k_1 = k_2$.

Example 27

Let K be a binomial ($n = 20, p = 0.4$) random variable. What is $P(K = 8)$?



Applications of the Central Limit Theorem

Solution. Since $E(K) = np = 8$ and $V(K) = np(1 - p) = 4.8$, the central limit theorem approximation to K is a normal random variable X with $E(X) = 8$ and $V(X) = 4.8$. Because X is a continuous random variable, $P(X = 8) = 0$, a useless approximation to $P(K = 8)$. On the other hand, the De Moivre–Laplace formula produces

$$P(8 \leq K \leq 8) \simeq P(7.5 \leq X \leq 8.5) = \Phi\left(\frac{0.5}{\sqrt{4.8}}\right) - \Phi\left(\frac{-0.5}{\sqrt{4.8}}\right) = 2\Phi(0.23) - 1 = 0.1819.$$

The exact value is $P(X = 8) = C_{20}^8(0.4)^8(0.6)^{12} \simeq 0.1797$.



Joint Cumulative Distribution Function

Example 28

K is the number of heads in 100 flips of a fair coin. What is $P(50 \leq K \leq 51)$?

Solution. Since K is a binomial ($n = 100, p = 1/2$) random variable,

$$P(50 \leq K \leq 51) = P(K = 50) + P(K = 51) = C_{100}^{50}(0, 5)^{100} + C_{100}^{51}(0, 5)^{100} \simeq 0.1576.$$

Since $E(K) = 50$ and $\sigma_K = 5$, the ordinary central limit theorem approximation produces

$$P(50 \leq K \leq 51) \simeq \Phi\left(\frac{51 - 50}{\sqrt{25}}\right) - \Phi\left(\frac{50 - 50}{\sqrt{25}}\right) = \Phi(0.33) - \Phi(0) = 0.62930 - 0.5 = 0.1293.$$

This approximation error of roughly 50% occurs because the ordinary central limit theorem approximation ignores the fact that the discrete random variable K has two probability masses in an interval of length 1. As we see next, the De Moivre–Laplace approximation is far more accurate.

$$\begin{aligned} P(50 \leq K \leq 51) &\simeq \Phi\left(\frac{51 + 0.5 - 50}{\sqrt{25}}\right) - \Phi\left(\frac{50 - 0.5 - 50}{\sqrt{25}}\right) = \Phi(0.3) - \Phi(0.1) \\ &= 0.61791 + 0.53983 - 1 = 0.1577. \end{aligned}$$



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Discrete Random Variables

Exercise 1

Random variables X and Y have the joint CDF

$$F_{X,Y}(x,y) = \begin{cases} (1 - e^{-x})(1 - e^{-y}), & x \geq 0, y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) What is $P(X \leq 2, Y \leq 3)$?
- (b) What is the marginal CDF, $F_X(x)$?
- (c) What is the marginal CDF, $F_Y(y)$?

Exercise 2

Random variables X and Y have CDF $F_X(x)$ and $F_Y(y)$. Is $F(x,y) = F_X(x)F_Y(y)$ a valid CDF? Explain your answer.



Discrete Random Variables

Exercise 3

Random variables X and Y have the joint PMF

$$P_{X,Y}(x,y) = \begin{cases} cxy, & x = 1, 2, 4, \ y = 1, 3, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (a) What is $P(Y < X)$?
- (b) What is $P(Y > X)$?
- (c) What is $P(Y = X)$?
- (d) What is $P(Y = 3)$?



Discrete Random Variables

Exercise 4

Random variables X and Y have the joint PMF

$$P_{X,Y}(x,y) = \begin{cases} c|x+y|, & x = -2, 0, 2, \ y = -1, 0, 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P(Y < X)$?
- (c) What is $P(Y > X)$?
- (d) What is $P(Y = X)$?
- (e) What is $P(X < 1)$?



Discrete Random Variables

Exercise 5

Given the random variables X and Y in Problem 4, find

- (a) The marginal PMFs $P_X(x)$ and $P_Y(y)$,
- (b) The expected values $E[X]$ and $E[Y]$,
- (c) The standard deviations σ_X and σ_Y .

Exercise 6

Given the random variables X and Y in Problem 5, find

- (a) The marginal PMFs $P_X(x)$ and $P_Y(y)$,
- (b) The expected values $E(X)$ and $E(Y)$,
- (c) The standard deviations σ_X and σ_Y .



Discrete Random Variables

Exercise 7

Random variables X and Y have the joint probability distribution

$X \backslash Y$	1	2	3
1	0.12	0.15	0.03
2	0.28	0.35	0.07

- (a) Are X and Y independent?
- (b) Find the marginal probability distribution of X and Y .
- (c) Find the probability distribution of Z , where $Z = XY$.
- (d) Find $E(Z)$. Proof $E(Z) = E(X) \cdot E(Y)$.



Discrete Random Variables

Exercise 8

Random variables X and Y have the joint probability distribution

$X \backslash Y$	-1	0	1
-1	$\frac{4}{15}$	$\frac{1}{15}$	$\frac{4}{15}$
0	$\frac{1}{15}$	$\frac{2}{15}$	$\frac{1}{15}$
1	0	$\frac{2}{15}$	0

- (a) Find $E(X)$, $E(Y)$, and $\text{cov}(X, Y)$.
- (b) Are X and Y independent?
- (c) Find the marginal probability distribution of X and Y .



Discrete Random Variables

Exercise 9

Random variables X and Y have the joint probability distribution

$X \backslash Y$	1	2	3
1	0.17	0.13	0.25
2	0.10	0.30	0.05

- Find the marginal probability distribution of X and Y .
- Find the covariance matrix of X and Y .
- Find the correlation coefficient of two random variables X and Y .
- Are X and Y independent?



Continuous Random Variables

Exercise 10

Random variables X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} c, & x + y \leq 1, \ x \geq 0, \ y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P(X \leq Y)$?
- (c) What is $P(X + Y \leq 1/2)$?



Continuous Random Variables

Exercise 11

Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} cxy^2, & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{otherwise} \end{cases}$$

- (a) Find the constant c .
- (b) Find $P(X > Y)$ and $P(Y < X^2)$.
- (c) Find $P[\min(X, Y) \leq 1/2]$.
- (d) Find $P[\max(X, Y) \leq 3/4]$.

Continuous Random Variables



Exercise 12

X and Y are random variables with the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 2, & x+y \leq 1, \ x \geq 0, \ y \geq 0, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) What is the marginal PDF $f_X(x)$?
- (b) What is the marginal PDF $f_Y(y)$?



Continuous Random Variables

Exercise 13

Random variables X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} cy, & 0 \leq y \leq x \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Draw the region of nonzero probability.
- (b) What is the value of the constant c ?
- (c) What is $F_X(x)$?
- (d) What is $F_Y(y)$?
- (e) What is $P(Y \leq X/2)$?



Continuous Random Variables

Exercise 14

Given random variables X and Y in Problem 5 and the function $W = X + 2Y$, find

- (a) The probability mass function $P_W(w)$,
- (b) The expected value $E(W)$,
- (c) $P(W > 0)$.

Exercise 15

Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} x+y, & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Let $W = \max(X, Y)$.

- (a) What is S_W , the range of W ?
- (b) Find $F_W(w)$ and $f_W(w)$.



Continuous Random Variables

Exercise 16

For the random variables X and Y in Problem 4, find

- (a) The expected value of $W = Y/X$,
- (b) The correlation, $E(XY)$,
- (c) The covariance, $\text{cov}(X, Y)$,
- (d) The correlation coefficient, ρ_{XY} ,
- (e) The variance of $X + Y$, $\text{Var}(X + Y)$.



Continuous Random Variables

Exercise 17

Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} (x+y)/3, & 0 \leq x \leq 1, 0 \leq y \leq 2, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) What are $E(X)$ and $\text{Var}(X)$?
- (b) What are $E(Y)$ and $\text{Var}(Y)$?
- (c) What is $\text{cov}(X, Y)$?
- (d) What is $E(X + Y)$?
- (e) What is $\text{Var}(X + Y)$?



Continuous Random Variables

Exercise 18

Random variables X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 5x^2/2, & -1 \leq x \leq 1, 0 \leq y \leq x^2, \\ 0, & \text{otherwise.} \end{cases}$$

Let $A = \{Y \leq 1/4\}$.

- (a) What is the conditional PDF $f_{X,Y|A}(x,y)$?
- (b) What is $f_{Y|A}(y)$?
- (c) What is $E(Y|A)$?
- (d) What is $f_{X|A}(x)$?
- (e) What is $E(X|A)$?



Continuous Random Variables

Exercise 19

X and Y have joint PDF

$$f_{X,Y}(x,y) = \begin{cases} (4x + 2y)/3, & 0 \leq x \leq 1, 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) For which values of y is $f_{X|Y}(x)$ defined? What is $f_{X|Y}(x)$?
- (b) For which values of x is $f_{Y|X}(y)$ defined? What is $f_{Y|X}(y)$?



Continuous Random Variables

Exercise 20

The joint PDF of two random variables X and Y is

$$f_{X,Y}(x,y) = \begin{cases} kx, & \text{if } 0 < y < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

- (a) Find the constant k .
- (b) Find the PDFs of X and Y .
- (c) Are X and Y independent?